

Data Analytics (DA)

Module -3

MySQL Assignment 2 -Querying Data – EXECUTION PDF

Dataset Description:

create the following three tables:

- **Table: Department**

Attributes:

- department_id – Primary Key
- department_name

- **Table: Location**

Attributes:

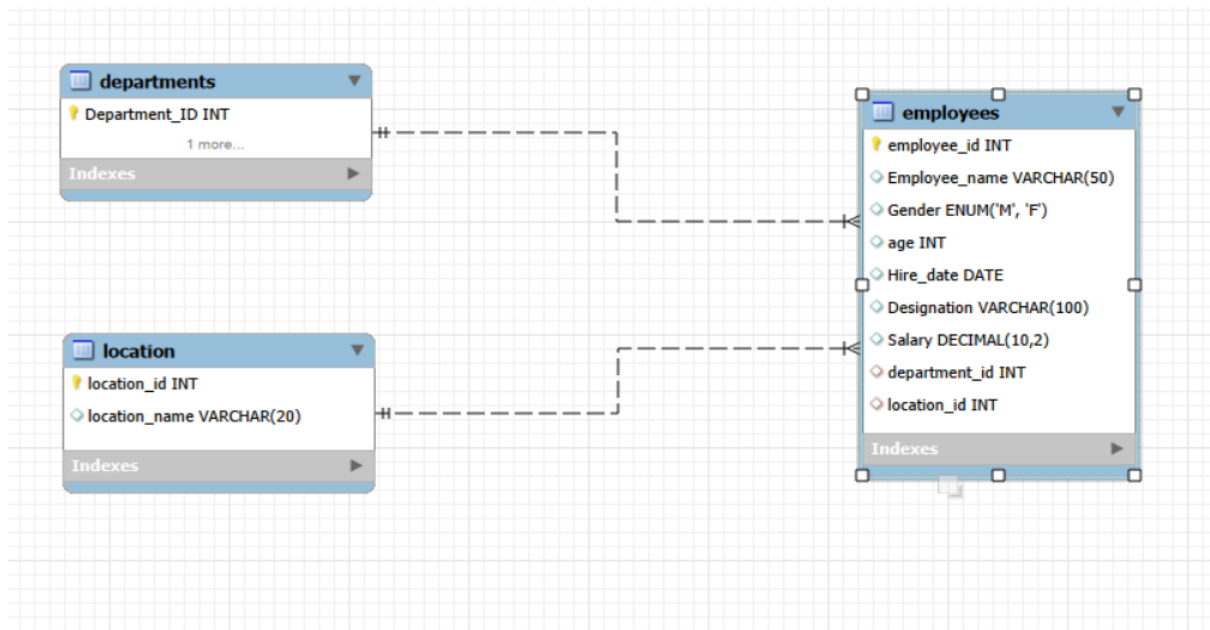
- location_id – Primary Key
- location_name

- **Table: Employees**

Attributes:

- employee_id – Primary Key
- Employee_name
- Gender
- Age
- Hire_date
- Designation
- Salary
- department_id – Foreign Key referencing department(department_id)
- location_id – Foreign Key referencing location(location_id)

Data Modelling :



Tasks

Clause & Operators:

1. DISTINCT VALUES:

* A query to retrieve distinct salaries from the Employees table.

```
57  /* 1. DISTINCT VALUES:
58  -- A query to retrieve distinct salaries from the Employees table.
59  */
```

```
61  • select distinct salary from employees ;
```

```
62
```

result Grid | Filter Rows: | Export: | Wrap Cell Content: |

salary
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

employees 5 x

2. ALIAS (AS):

* Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
63  /* * Provide aliases for the "age" and "salary" columns as "Employee_Age" and
64  "Employee_Salary", respectively.
65  */
66
67  • select
68      age as Employee_Age , Salary as Employee_Salary
69      from employees ;
70
```

Result Grid

	Employee_Age	Employee_Salary
▶	27	60000.00
	31	80000.00
	26	45000.00
	29	70000.00
	25	55000.00
	28	65000.00
	27	60000.00

employees 6 x

Output

#	Time	Action	Message
✓	10 20:24:28	CREATE TABLE Employees (employee_id INT PRIMARY KEY, employee_name VARCHAR(50) NO...	0 row(s) affected
✓	11 20:24:54	INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, department...	30 row(s) affected Records: 30 Duplicates: 0 \
✓	12 20:25:01	select distinct salary from employees LIMIT 0, 1000	11 row(s) returned
✓	13 20:32:19	select distinct salary from employees LIMIT 0, 1000	11 row(s) returned
✓	14 20:34:46	select age as Employee_Age , Salary as Employee_Salary from employees LIMIT 0, 1000	30 row(s) returned

3. WHERE CLAUSE & OPERATORS:

* Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
71  /* * Retrieve employees with a salary greater than ₹50000 and hired before
72  2016-01-01.
73  */
74
75  • select
76      Employee_name , Salary , Hire_date
77      from employees
78      where salary > 50000 and hire_date < '2016-01-01' ;
79
80
```

Result Grid

	Employee_name	Salary	Hire_date
▶	Vihaan Singh	60000.00	2015-01-20
	Reyansh Singh	80000.00	2015-03-10
	Kara Malhotra	70000.00	2015-07-05
	Anvi Chaudhary	55000.00	2015-09-11
	Dhruv Shetty	65000.00	2015-11-20

employees 8 x

Output

#	Time	Action	Message
✓	12 20:25:01	select distinct salary from employees LIMIT 0, 1000	11 row(s) returned
✓	13 20:32:19	select distinct salary from employees LIMIT 0, 1000	11 row(s) returned
✓	14 20:34:46	select age as Employee_Age , Salary as Employee_Salary from employees LIMIT 0, 1000	30 row(s) returned
✓	15 20:38:46	select Employee_name , Salary , Hire_date from employees where salary > 50000 and hire_date ...	0 row(s) returned
✓	16 20:46:47	select Employee_name , Salary , Hire_date from employees where salary > 50000 and hire_date ...	5 row(s) returned

*** Find the employee whose designation is missing and fill it with "Data Scientist".**

```

1
2 • select
3     employee_id , employee_name , Designation
4     from employees
5     where designation is null ;
6
7
8 •
9     update employees
10    set designation = 'Data Scientist'
11    where designation is null ;
12
13
14 • select *
15     from employees
16     where designation = 'data Scientist' ;
17
18

```

employee_id	employee_name	gender	age	hire_date	designation	salary	department_id	location_id
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	70000.00	8	3
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	100000.00	3	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. Sorting and Grouping Data:

1. ORDER BY:

• Find employees sorted by department ID in ascending order and salary in descending order.

```

95
96 /* Find employees sorted by department ID in ascending order and salary in
97    descending order.
98 */
99
100 • select
101     employee_name , department_id , salary
102     from employees
103     order by department_id asc , salary desc
104     ;
105
106
107

```

employee_name	department_id	salary
Aarav Nair	1	90000.00
Anushka Singh	2	90000.00
Anika Paul	2	70000.00
Saanvi Patel	2	60000.00
Avni Iyengar	3	100000.00

employees 13 x

2. LIMIT:

- Display the first 5 employees hired in the year 2018.

```
100
107  /* Display the first 5 employees hired in the year 2018 */
108
109 • select
110     employee_name, hire_date
111     from employees
112     where hire_date between '2018-01-01' and '2018-12-01'
113     order by hire_date asc
114     limit 5 ;
115
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch

	employee_name	hire_date
▶	Ishika Patel	2018-01-15
	Aarav Nair	2018-03-05
	Advik Kapoor	2018-05-20
	Aadhya Iyengar	2018-07-10
	Anika Paul	2018-09-20

employees 19 x

3. AGGREGATE FUNCTIONS:

- Calculate the sum of all salaries in the Finance department.

```
118 • select
119     sum(salary) as total_salary
120     from employees
121     where department_id = 7 ;
122
123
124
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	total_salary
▶	170000.00

- Find the minimum age among all employees.

```
124
125 • select
126     min(age) as min_age
127     from employees ;
128
129
130
```

Result Grid | Filter Rows: | Export

	min_age
▶	25

4. GROUP BY:

- List the maximum salary for each location.

```
128
129 /* List the maximum salary for each location */
130
131 • select
132     Location_id ,
133     max(salary) as Max_salary
134     from employees
135     group by location_id ;
136
137
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

	Location_id	Max_salary
▶	4	100000.00
	1	95000.00
	2	95000.00
	3	90000.00

- Calculate the average salary for each designation containing the word 'Analyst'.

```

137  /* Calculate the average salary for each designation containing the word 'Analyst' */
138
139  • select
140      designation ,
141      avg(salary) as avg_salary
142  from employees
143  where designation like '%Analyst%'
144  group by designation ;
145
146
147

```

Result Grid

designation	avg_salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000

5. HAVING:

- Find departments with less than 3 employees.

```

146
147  /* Find departments with less than 3 employees */
148
149  • select
150      department_id ,
151      count(*) as total_employees
152  from employees
153  group by department_id
154  Having total_employees < 3 ;
155
156
157
158

```

Result Grid

department_id	total_employees
7	2
9	1
1	1
5	2
6	2

- Find locations with female employees whose average age is below 30.

```

156
157  /* Find locations with female employees whose average age is below 30 */
158
159 • select
160     Location_id , gender ,
161     avg(age) as avg_age
162     from employees
163     group by location_id ,Gender
164     having avg_age < 30 and gender = 'F';
165
166

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Location_id	gender	avg_age
▶	2	F	27.3333
	1	F	26.7500
	4	F	29.2000

Joins:

1. INNER JOIN:

- List employee names, their designations, and department names where employees are assigned to a department.

```

172 • select
173     e.employee_name ,
174     e.designation ,
175     d.department_name
176     from
177     employees as e
178     Inner join
179     departments as d
180     On
181     e.department_id = d.department_id ;
182
183

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	employee_name	designation	department_name
	Advik Kapoor	Finance Analyst	Finance
	Aadhya Iyengar	HR Specialist	Human Resources
	Anika Paul	Public Relations Specialist	Marketing
	Aryan Shetty	Product Manager	Product Management
	Avni Iyengar	Data Scientist	Data Science
	Vivaan Singh	Business Analyst	Data Science
	Ananya Paul	Content Writer	Content Creation
10)	Anaya Kapoor	Event Coordinator	Content Creation
	Arjun Kumar	Quality Assurance Analyst	IT
	Sara Iyer	Project Manager	Product Management

2. LEFT JOIN:

- List all departments along with the total number of employees in each department, including departments with no employees.

```
187
188 • select
189     d. department_name,
190     count(e.employee_id) as total_employee_count
191 from departments as d
192 left join
193 employees as e
194 on
195     d.department_id = e.department_id
196 group by
197     d.department_name , d.department_id ;
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	department_name	total_employee_count
▶	Business Development	2
	Content Creation	2
	Customer Support	3
	Data Science	3
	Design	3
	Finance	2
	Human Resources	3
	IT	4

Result 36 x

3. RIGHT JOIN:

- Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```

204 • select
205     e. employee_name ,
206     l. location_name
207 from
208     employees as e
209 right join
210     location as l
211 On
212     e.location_id = l.location_id ;
213
214

```

employee_name	location_name
Arjun Kumar	Bangalore
Vivaan Singh	Bangalore
Anika Paul	Bangalore
Kabir Nair	Bangalore
Vihaan Mohan	Bangalore
Myra Verma	Bangalore
Dhruv Shetty	Bangalore
Aaradhya Iyer	Bangalore
Sara Iyer	Chennai
Arjun Kumar	Chennai

Result 37

4. CROSS JOIN

- Show all possible combinations of departments and locations.

```

218 • select
219     d.department_name,
220     l. location_name
221 from
222     departments as d
223 cross join
224     Location as l ;
225
226 • select
227     d. department_id ,
228     l.location_Name
229 from
230     departments as d

```

department_name	location_name
Business Development	Pune
Business Development	Hyderabad
Business Development	Chennai
Business Development	Bangalore
Content Creation	Pune
Content Creation	Hyderabad
Content Creation	Chennai
Content Creation	Bangalore
Customer Support	Pune
Customer Support	Hyderabad

Result 44

5. SELF JOIN:

- Show pairs of employees working in the same department, excluding self-pairs.

```
247 • select
248     e1.employee_id ,
249     e2.employee_id ,
250     e1. department_id
251 from
252     employees as e1
253 join
254     employees as e2
255 On
256     e1. department_id = e2.department_id
257 And
258     e1.employee_id <> e2.employee_id ;
259
260
```

Result Grid			
Filter Rows:			
Export:  Wrap			
	employee_1	employee_2	department_id
	5025	5001	3
	5029	5002	12
	5017	5002	12
	5010	5002	12
	5014	5003	10
	5013	5003	10
	5008	5004	8
	5006	5004	8
	5009	5005	11
	5008	5006	8

Windows function

- Write a window function query to rank employees by salary using rank().

```
166
167  /* Write a window function query to rank employees by salary using rank() */
168
169 • select
170     employee_name , salary ,
171     rank() over ( order by salary desc) as rank_by_salary
172     from employees ;
173
174
175
```

Result Grid

	employee_name	salary	rank_by_salary
▶	Zoya Khan	100000.00	1
	Avni Iyengar	100000.00	1
	Myra Verma	95000.00	3
	Aryan Shetty	95000.00	3
	Anushka Singh	90000.00	5

- Write a window function query to rank employees by salary within each department using DENSE_RANK()

```
173
174  /* Write a window function query to rank employees by salary within each department
175     using DENSE_RANK()
176     */
177
178 • select
179     Employee_name , Salary , Department_id,
180     dense_rank() over ( partition by department_id order by salary desc ) as deptwise_ranking_by_salary
181     from employees ;
182
183
184
```

Result Grid

	Employee_name	Salary	Department_id	deptwise_ranking_by_salary
▶	Aarav Nair	90000.00	1	1
	Anushka Singh	90000.00	2	1
	Anika Paul	70000.00	2	2
	Saanvi Patel	60000.00	2	3
	Avni Iyengar	100000.00	3	1
	Vivaan Singh	75000.00	3	2
	Vihaan Singh	60000.00	3	3
	Zoya Khan	100000.00	4	1
	Aadhya Iyengar	60000.00	4	2
	Rishi Patel	55000.00	4	3

- Write a window function query, Running total salary by department

```
185
186 • select
187     employee_name , department_id ,salary ,
188     sum(salary) over (partition by department_id order by salary desc) as Running_total_salary
189     from employees ;
190
191
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	employee_name	department_id	salary	Running_total_salary
▶	Aarav Nair	1	90000.00	90000.00
	Anushka Singh	2	90000.00	90000.00
	Anika Paul	2	70000.00	160000.00
	Saanvi Patel	2	60000.00	220000.00
	Avni Iyengar	3	100000.00	100000.00
	Vivaan Singh	3	75000.00	175000.00
	Vihaan Singh	3	60000.00	235000.00
	Zoya Khan	4	100000.00	100000.00
	Aadhya Iyengar	4	60000.00	160000.00
	Ishika Patel	4	55000.00	215000.00

Result 34 ×